

Gnome Venture Style Sheet & Art Guide

Hearthlight art, inline icons, the UI rules and Aseprite pixel art

Game version 2.0.67
Built 27 October 2026
273 inline icons · 142 colour tokens · 54 pages

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About this guide

This is the one reference for how Gnome Venture looks: the Hearthlight art style, the inline icon system, and the UI style sheet. It is **rebuilt from the real files**, so the icon pages and colour tables are never out of date.

How to read it

- **Part A**: the art rules for every picture (placeholders, creatures, folk, nav).
- **Part B**: inline icons (`ico-*`): how emoji are swapped for our own drawings, and how to add one.
- **Part C**: the UI style sheet: colours, type, spacing, components, and Joshua's UI rules.
- **Part D**: Aseprite pixel art (Stitched Toy): the signature, canvas sizes, the 1 px black border, palettes, the trek map tiles, the Aseprite workflow and a per-asset checklist.

Rebuild it

```
python3 tests/build_guide.py
```

Output: `game/assets/guides/Gnome-Venture-Style-Guide.pdf` . Download it in Admin, Images & art, Art & style guide. Run it whenever an icon, a colour token or a rule changes. Text lives in `tests/guide_src/text.py` ; layout in `guide.css` ; component demos in `components.html` .

Where the facts come from

- Art rules: written from `tests/hearth.py` and `hearth_all.py` (constants and docstrings), `claude/handoff.md` and `assets/img/ph/palette.json` . This text also builds `claude/art-style-guide.md` (`tests/build_guide_md.py`); no older hand-written guide was found. Change `guide_src/text.py` to change both.
- Icons: `assets/img/ico/` and `includes/data/ico_map.php` , read at build time.
- UI: `assets/style.css` (142 tokens read), plus real screenshots of the component classes.

Hearthlight: the look

> **PNG art (v2.0.34):** the Aseprite PNGs of these pictures have **no cream halo**; they carry a 1 px black border instead (Part D). This part describes the SVG files, which still ship as the fallback and keep the halo and the layers below.

Hearthlight is the house style of every placeholder picture, creature, folk portrait, nav icon and inline icon. One look, three layers, drawn on a **64 × 64 grid**. All widths below are in grid units **u** (u = shorter side / 64; so 1 u = 1 px at 64 px). Pen weights below are for large stickers (pictures, creatures, portraits). **Icons (`ico-` names) use heavier pens**: see the icon rules.

Note: these rules are written from `tests/hearth.py`, `hearth_all.py` and `claude/handoff.md`. Text source for this PDF and for `claude/art-style-guide.md` is this file; no older hand-written guide was found.

The three layers

1. **Cream halo**: a thick sticker edge round the whole silhouette, `#f2e4c0`, 3.2 u out from the shape (stroke 6.4) on large stickers; icons use 6.0 u. It lifts the picture off dark and light pages alike.
2. **Umber line**: a warm near-black outline, `#2a1a10`, 1.9 u out from the shape (stroke 3.8) on large stickers; icons use 3.4 u. Never pure black.
3. **Cel shade**: one crescent at the lower right, clipped to the silhouette, in a dark tint of the picture's main colour, opacity .34. One light direction for every picture.

Halo: icons 6.0 u, stickers 3.2 u
`#f2e4c0`

Outline: icons 3.4 u, stickers 1.9 u
`#2a1a10`

Shade crescent
`#17361c`

Small rules

- Eyes: umber pupil (at least 1.25 u radius) on a cream disc, with a small cream glint on big pupils. A dark eye on a dark fill gets a cream disc and a ring so it reads.
- Inner contour: 1.0 u umber line at 55% opacity between big shapes. Dropped first when a file runs over its size cap.
- Wood and shafts keep body: no wooden line thinner than 6.6 u wide.
- Hairlines (legs, feelers, whiskers) snap to 1 or 1.5 u. Under 1 u they vanish at 16 px.
- Folk portraits (`npc-*`) use 60% halo and outline, because the portrait frame crops them.
- Nav icons (24 px) use whole-number coordinates, and rounded corners under 3 u are dropped.
- Ground shadow: umber at 42% opacity, an ellipse under standing things. Inline icons have none.

Palette

Every picture starts from the shared art palette (`assets/img/ph/palette.json`). The restyle then nudges a few cold steps warmer (table below). Shade = the next darker step of the same family.

umber line
#2a1a10

cream halo
#f2e4c0

rim light
#f0d49a

rim tone
#fff1c8

Family	Steps (step number, then colour)
soil	900 700 500 300 100
moss	800 600 400 200
gold	700 500 300
ember	700 500 400 300
petal	500 300
water	700 500 300
dusk	700 500 300
stone	700 500 300
ink	#2b2b2b
cream	#f4f0e0
placeholder_outline	#e0392b

Warmth nudges (old to new)

Picture colour	Becomes
#2b2b2b	#2a1a10
#000	#2a1a10
#000000	#2a1a10
#7a4a1f	#8c4a1a
#9a6b3f	#a5663a
#3d2a1c	#44281c
#c9a96e	#dcb06a
#8b8f96	#968f96
#5b6470	#6a6070
#c9c5bc	#d6c9b6
#4f9a45	#58a03e
#2f5a2a	#2e5a2c
#a4b86a	#b0bc66
#25637d	#256a80
#3f8fb0	#3f93b2

Creature / folk colour	Becomes
#6a6070	#8f7060
#968f96	#a38f80
#d6c9b6	#e6d2ae
#e46a9b	#d9667f
#f4f0e0	#f8eed2
#f1ece0	#f8eed2
#a9d6ea	#b8d8d2
#a8241f	#9c2a1e

Shade tints

The cel crescent is not black. It takes the tint of the picture's dominant hue.

greens
#17361c

reds
#5a1410

blues
#12324a

purples
#2e1a5a

golds, yellows
#5a3008

browns, oranges
#4a1c10

greys
#3a2236

Limits, checks and new art

Specimens

Existing Hearthlight pictures from `assets/img/ph/`, shown at 96 px.


tomato


sunflower


acai


c-bumblebee


c-anteater


npc-wren


nav-garden


sunflower-snack

Size caps

Every SVG has a byte budget. The restyle walks down a **trim ladder** until a file fits.

Kind	Hard cap	Notes
Most pictures, icons, items	4096 B	Ladder aims at 3500 B
`creature-*`, `npc-*`	4096 B	Lean mode (1 decimal, no rim). Over 3900 B: swarm mode
`banner-*`, `land-*`, `guide-*`	8192 B	Scenes: full-bleed allowed

Trim ladder (richest first)

1. Full: rim light, ambient shade, glint, inner contour, material marks.
2. No rim / AO / glint.
3. Also no inner contour.
4. Also no material marks.
5. Capsule silhouettes: ellipses become round-cap strokes, small corner radii vanish.
6. Capsules, no inner contour.
7. Capsules, no inner contour, no marks.

Checks every file must pass

- No `<script>`, `<style>`, `style=`, filters, gradients, `xlink`, `href`, `<image>`, `<foreignObject>`, `<use>` or `on*=` handlers.

- `viewBox` unchanged by the restyle.
- Every `clipPath` id is unique (`hl-<name>`), so two icons on one page never clash.
- Rendered at 64 px, the halo must **not touch the 1 px frame** and the picture must fill at least 3% of it. Scenes, tiles, fences and `npc-*` busts are exempt (they bleed by design).
- Names: lowercase, hyphens. Prefixes mean something: `creature-` , `npc-` , `nav-` , `banner-` , `land-` , `guide-` , `tile-` .

New or changed pictures

1. Draw plain SVG (no effects) in the game palette.
2. Restyle **last**: `PYTHONHASHSEED=1 python3 -B tests/hearth_all.py game/assets/img/ph <out>`
3. Check edges: `python3 -B tests/hearth_edge.py <out>` (0 flagged).
4. Copy `<out>/*.svg` over `game/assets/img/ph/` . Never copy `palette.json` .
5. **Never** re-run `gen_ph_*.py` over restyled files. It would undo the style.

Known weak spots

Ember beetle eyes are hard to read. Nav festival, friends, villagers and arranging mush together at 24 px. Nav account has low contrast on dark.

Inline icons: how they work

Inline icons replace emoji **inside page text** with our own Hearthlight drawings. Same look everywhere, no phone-to-phone emoji differences.

How the swap works

- It happens at **render time**. `includes/emoji_icons.php` runs as the page's output buffer (`ico_start()` in bootstrap) and swaps emoji for `<img class="ico">`.
- The map is `includes/data/ico_map.php`: emoji to slug, decorative flag, accessible name. Keys have no U+FE0F.
- Files live in `assets/img/ico/{slug}.svg`. **No file = the emoji stays.** So a map row can be added before the art.
- Longest match wins (a flag or family sequence beats its first emoji).
- Only text nodes are swapped. Never attributes, `<option>`, `<title>`, `<select>`, `<textarea>`, `<script>`, `<style>`, `<svg>`, `<noscript>`.
- JSON, CSV, XML, redirects and downloads are skipped. Setting `ico_swap` (default on) turns it off.
- The icon is an `<img>`: copy and paste of the page gives nothing for it.

Player text keeps its emoji

Anything a player typed must print with `eu()` (emoji become `&#x..;` and the swap skips them): garden names, critter names, trade and job notes, carving titles, feedback, report details.

Known limits

Emoji inside JS-built strings, attributes, `<option>`, `<title>` and SVG text stay emoji. Player names or notes copied into notices or flash messages (stored as plain text) can get swapped.

Decorative or named?

- **Decorative** (most): `alt=""`. The words next to it carry the meaning.
- **Named**: the icon stands alone (energy, Petals, locked, done). Give it an accessible name in the map row.

Icon size and grid

The grid

Every icon is a **64 × 64** `viewBox` and is drawn as a plain silhouette first. Keep shapes chunky: it must read at **16 px**.

- Keep the artwork at least **6.5 u** away from the frame. The halo (6.0 u for icons) plus anti-alias must stay inside.
- One clear shape, 2 or 3 colours from the palette, one crescent of shade. No fine detail under 1 u.
- Pen weights for icons (the restyle adds them; `ICO_HALO` / `ICO_OUT` in `tests/hearth.py`): **halo 6.0 u, outline 3.4 u**, so the cream ring sits 2.6 u beyond the outline. Large stickers keep 3.2 / 1.9. Icons are drawn as plain shapes: the build adds the pens. (These are the SVG pens; the PNG icons have no halo, see Part D.)
- Size cap: 4096 bytes per icon. Typical is 1.5 to 3.5 KB.

Sizes in the game

Icons are sized in **em**, so they follow the text they sit in. At body size that is 16 to 18 px. Allowed range inline: **16 to 24 px**. Bigger art (48 px and up) is not an inline icon: use a placeholder picture.

Where the icon sits	Size
`.ico`	1.1em
`h1 .ico,h2 .ico,h3 .ico`	1em
`button .ico,.btn .ico,a.btn .ico`	1.05em
`nav.tabs .ico,nav.chips .ico,.tab .ico,.chip .ico`	1.05em
`.choice .ico`	1.1em
`.choice-ico .ico`	1em
`.choice-badge .ico,.choice-sub .ico`	1em
`summary .ico`	1.05em
`small .ico,.muted .ico`	1em
`.statusstrip .ico,.userbar .ico`	1.1em

Seen at size



16 px 20 px 24 px 48 px

Watering · Dark chip

Watering · Light chip

Rules for placing icons

- Icon first, then words. Never an icon alone unless it has an accessible name.
- One icon per label. Do not stack two.

- Keep the same icon for the same thing everywhere (the map guarantees it).
- Do not recolour with CSS. The icon carries its own colours and halo.

Adding an icon

Add an icon in 5 steps

1. **Draw** plain SVG on the 64 grid in `tests/ico_src/ico-<slug>.svg`. Flat fills from the palette. No gradients, text or filters.
2. **Map**: add one row to `includes/data/ico_map.php`:

   ```
   '🐝' => ['ico-bee', true],
   '⚡' => ['ico-energy', false, 'Energy'],
   ```

   The second value is “decorative” (true = no spoken name). Use false and a name when the icon stands alone.
3. **Restyle**: `PYTHONHASHSEED=1 python3 tests/ico_build.py restyle`. This writes `assets/img/ico/ico-<slug>.svg`. Never edit that folder by hand.
4. **Check**: `python3 -B tests/hearth_edge.py game/assets/img/ico` (0 flagged) and `php tests/emoji_icons.php game`.
5. **Regenerate this guide**: `python3 tests/build_guide.py`, then upload `assets/guides/` with the release.

Naming

- Files: `ico-<slug>.svg`, lowercase, hyphens: `ico-watering-can.svg`.
- Name the **thing**, not the emoji: `ico-gnome-hat`, not `ico-pointy-hat-face`.
- One slug can serve several emoji (variants). Do not make near-duplicates.
- Prefixes `creature-`, `npc-`, `nav-` are for pictures, not inline icons.

Do and don’t

Do

- ✓ Draw plain SVG on the 64 grid
- ✓ Keep 6.5 u clear of the frame
- ✓ Make it read at 16 px
- ✓ Use palette colours
- ✓ Run the restyle, then the edge check
- ✓ One map row per emoji
- ✓ Give a name to icons that stand alone
- ✓ Print player text with `eu()`

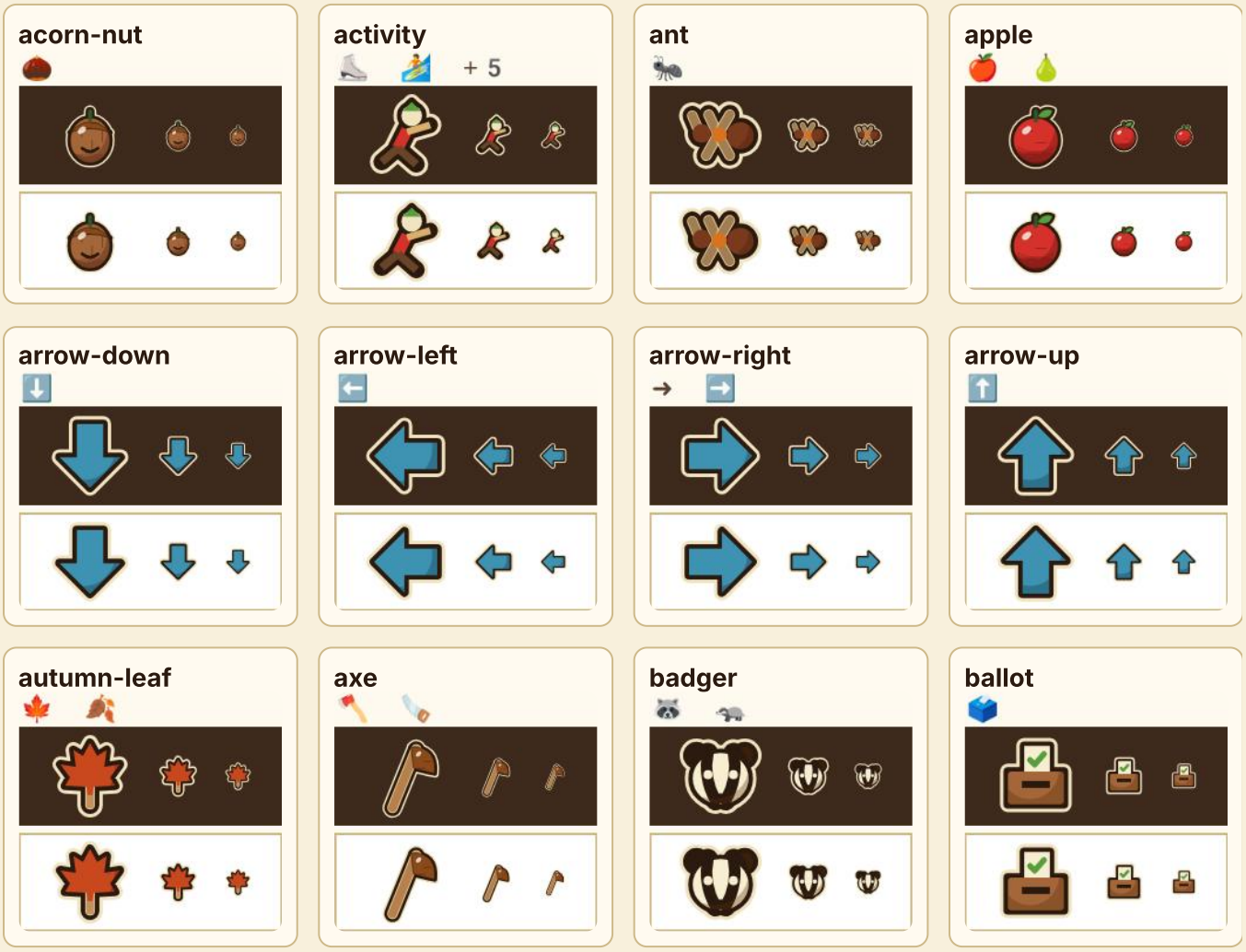
Don’t

- ✗ Gradients, text, filters, `<style>`, scripts
- ✗ Strokes under 1 u
- ✗ Hand-edit files in `assets/img/ico/`
- ✗ Recolour an icon with CSS
- ✗ Put emoji in new UI text
- ✗ Rely on colour alone for meaning
- ✗ Expect swaps in attributes, options, titles or JS strings
- ✗ Re-run `gen_ph_*.py` over restyled art

Icon catalogue

Everything in `assets/img/ico/`, read when this guide was built. Each card: the icon name, the emoji it replaces (from `ico_map.php`), then the icon at **48, 24 and 16 px** on a dark chip (the game's dark surface) and a light chip.

Check	Result
Icons in the folder	273
With an emoji map row	273 (the rest still show their emoji until mapped)
Total size / largest	461 KB / clap (3926 B)
Over the 4096 B cap	0
Banned content or odd viewBox	0
Duplicate internal ids	0



banana + 1



barrel



basket



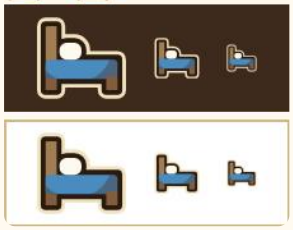
bat



bear



bed



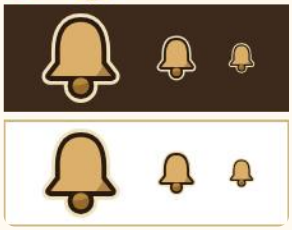
bee



beetle



bell



bird + 3



blossom + 4



boat + 1



bone



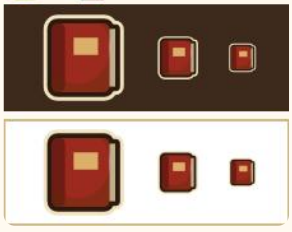
book-green



book-open



book-red



books



boot



bouquet



bow



box



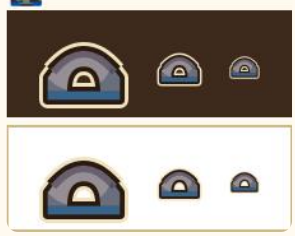
brain



bread + 2



bridge



bucket



bulb



burst



butterfly



cake + 1



calendar



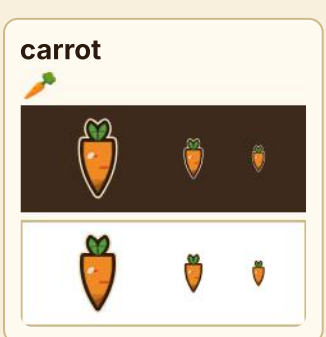
candy



carp-streamer



carrot



castle



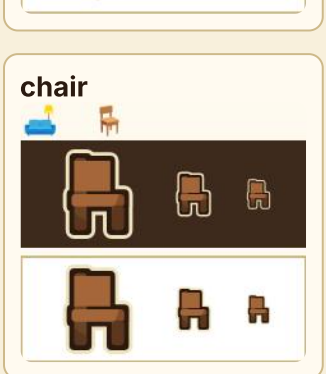
cat + 1



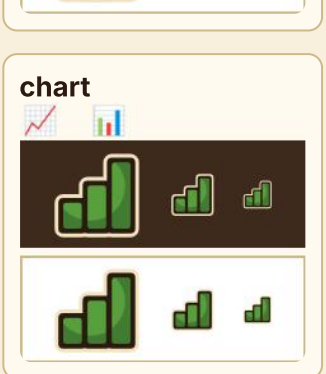
caterpillar



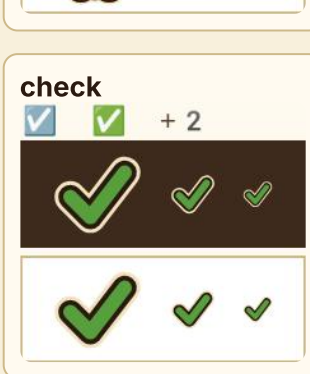
chair



chart



check + 2



cherry



chicken



chili



circus-tent



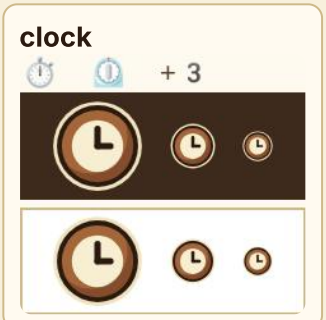
clap



clipboard



clock



cloud



clover



coffee



cog



coin



compass



compost



cooking-pan



corn



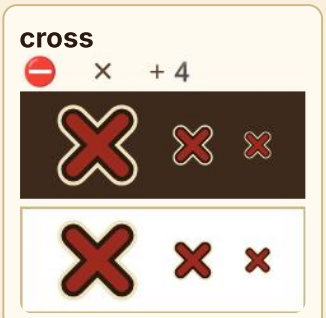
crab



cricket



cross



crown



crystal-ball



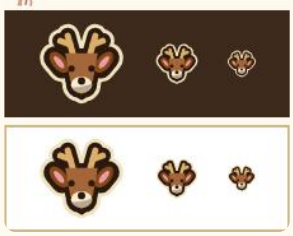
curry



daisy



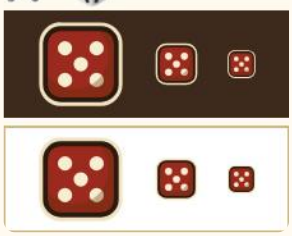
deer



desert



dice



dna



door



dove



dragon



droplet + 1



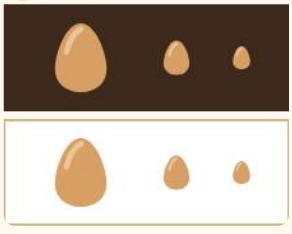
drum



duck



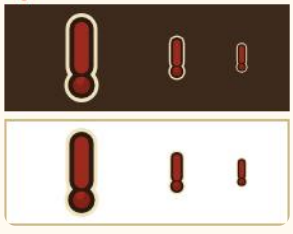
egg



envelope



exclaim



eyes



face-sigh + 3



face-smile + 2



feather



finish



fire



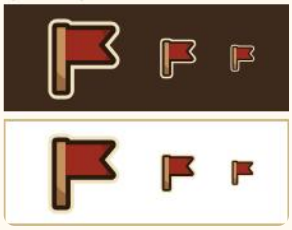
fish



fist



flag-red



flatbread



floppy



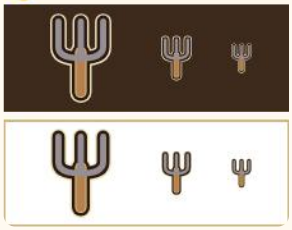
fly



fog



fork



fountain



four-leaf-clover



fox



frog



gem



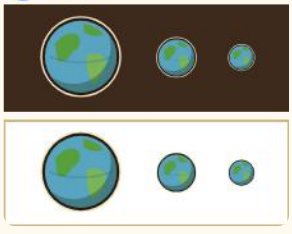
ghost



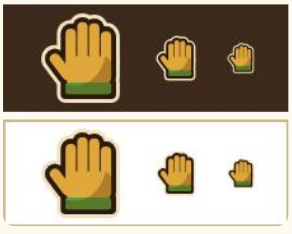
gift



globe



gloves



gnome-gardener



gnome-hat



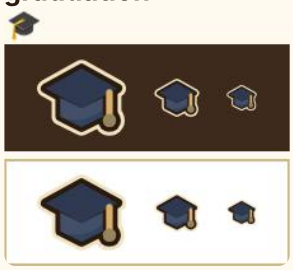
goat



goblin



graduation



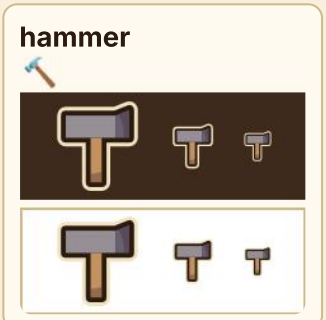
grain



grapes



hammer



hand-stop



hand-wave



handbag



heart-green



heart-red



heart-yellow



hedgehog



herb-sprig



hole



home



honey-pot



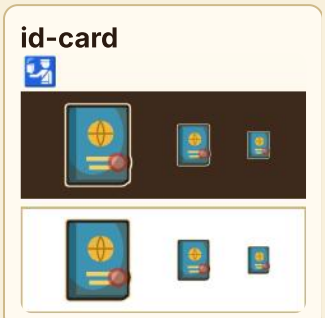
hourglass



house-row



id-card



info



island



jar



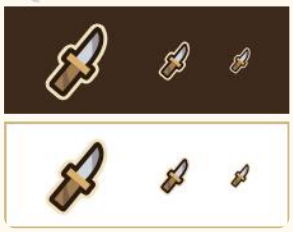
juice



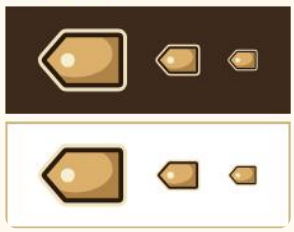
key



knife



label-tag



ladder



ladybird



lantern



leaf



leafy-green



lightning



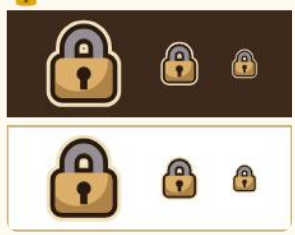
link



lizard



lock



log



magnifier



mailbox



map-pin



medal-bronze



medal-gold



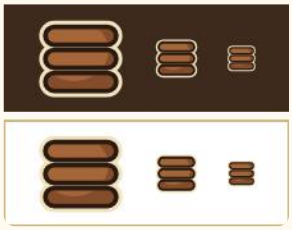
medal-silver



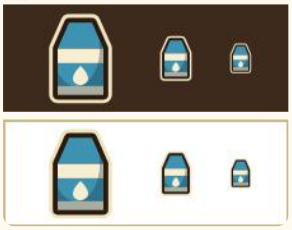
melon



menu



milk



money-bag



monkey



moon



moss-rock



mountain



mouse



mushroom



music



newspaper



notepad



olive



onion



orange



otter



owl



page



palette



pancakes



party



path



paw



peach

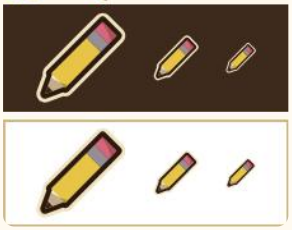


peanut



pencil

+ 1



people



person

+ 5



pick



pie

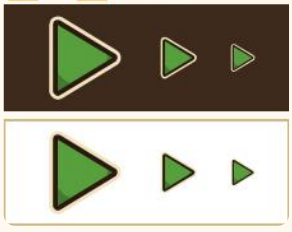


plate-cutlery



play

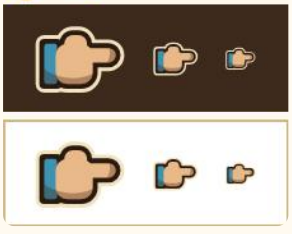
+ 1



plus



pointing



porridge



potato



pudding



pumpkin



pushpin



puzzle



question



rabbit



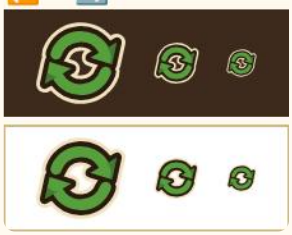
rain



rainbow



refresh



ribbon



roadworks



roast



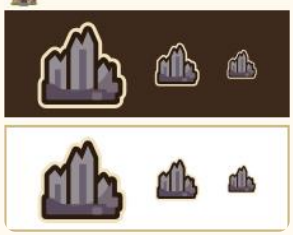
rod



rope



ruin



ruler



safari



salad



satchel



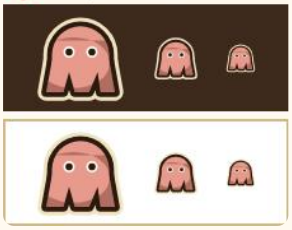
scales



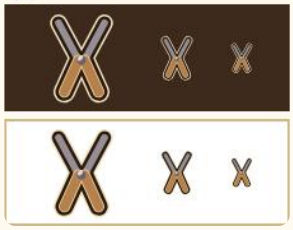
scroll



sea-creature + 1



shears



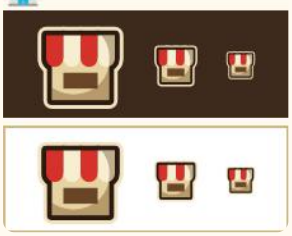
shell + 1



shield



shop



shrine



shuffle



signpost



skull



snail



snake



snowflake + 1



soup-bowl + 3



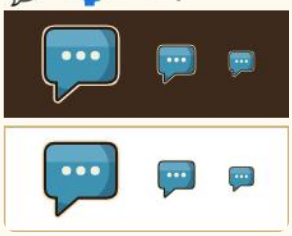
spade



sparkle



speech + 1



spider



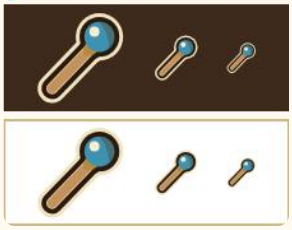
sprout + 1



squirrel



staff



stamp-cart



star-empty



star

+ 3



stone-wall



storm



strawberry



strength



sun-rain



sun

+ 4



sunflower



sword



taco

+ 1



target



tea



telescope



tent



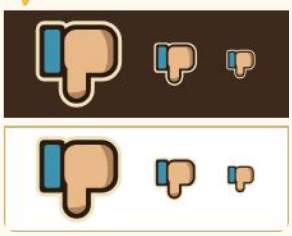
test-tube



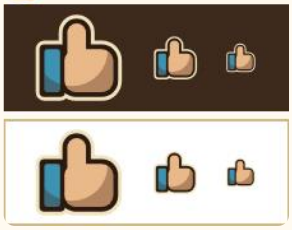
thermometer



thumbs-down



thumbs-up



tomato



toolbox



tools



tower



tree



trophy



trowel-pot



turtle



undo



walker



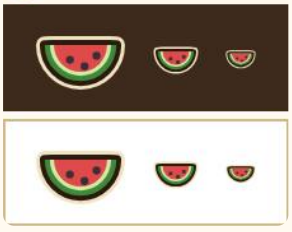
warning



watering-can



watermelon

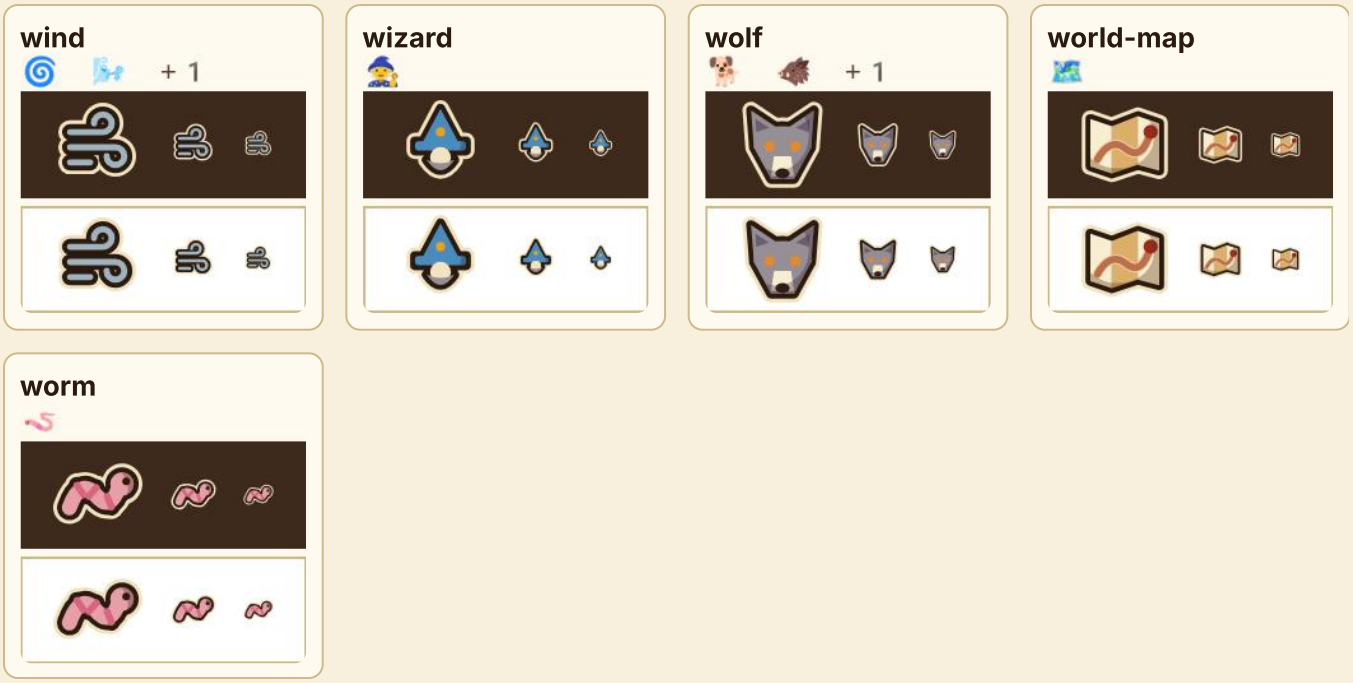


wave



wind-chime





Map rows without a file yet

These emoji still show as emoji until the icon is drawn (step 1 of “Adding an icon”).

2 to draw: `ico-handshake` `ico-prayer`

Colour tokens

All colours are **tokens** in `assets/style.css`, section 1 (`:root`). Components never hard-code a colour; dark mode only changes the tokens. Dark is the default. Dark overrides are always written `:root[data-theme="dark"] X`, never `@media (prefers-color-scheme)`.

Since v2.0.32 (**Stitched Toy chrome**) every token is a colour from the art palette (Part A families, the Part D table, or the Part D hue-shift recipe, marked in a CSS comment): cream and umber felt, moss for the main verb, gold thread for stitches. No pure black or white. The header, phone bottom bar and footer are umber felt (`--felt`) in both themes.

Surfaces and text

Light	Dark	Token	Light hex	Dark hex	Use
		`--bg`	#f1ece0	`#2a1a10`	Page background
		`--surface`	#fff3c8	`#3d2a1c`	Cards, inputs, popups
		`--surface-2`	#f2e4c0	`#4a2c1a`	Quiet fills: tabs, hover, strips
		`--box-bg`	#fff3c8	`#3d2a1c`	Card fill
		`--ink`	#2a1a10	`#f2e4c0`	Main text
		`--ink-2`	#4a2c1a	`#c9c5bc`	Second text
		`--muted`	#74462a	`#c9a96e`	Hints, help lines
		`--line`	#c9a96e	`#74462a`	Light frame edge, rules
		`--line-2`	#9a6b3f	`#9a6b3f`	Input borders, dark frame edge

Moss (brand)

Light	Dark	Token	Light hex	Dark hex	Use
		`--brand`	#2f5a2a	`#7ab55c`	Moss: links, good accents
		`--brand-ink`	#2f5a2a	`#a4b86a`	Green text
		`--brand-soft`	#c8d09f	`#1f3f3a`	Green wash (good, selected)
		`--brand-line`	#7ab55c	`#2f6a3a`	Green border
		`--brand-fill`	#2f5a2a	`#2f5a2a`	Solid moss behind cream text
		`--badge-bg`	#4a2c1a	`#74462a`	Solid umber behind cream text

Gold, red, blue, purple

Light	Dark	Token	Light hex	Dark hex	Use
		`--accent`	#e8a317	`#e8a317`	Gold: energy, notices, highlights
		`--accent-soft`	#fff3c8	`#6a4510`	Gold wash
		`--accent-ink`	#7a4a1f	`#f5cf6a`	Gold text
		`--accent-line`	#e8a317	`#b07a10`	Gold border
		`--danger`	#a8241f	`#f08a24`	Red: errors, destructive
		`--danger-soft`	#f6b8cf	`#6b1b2c`	Red wash
		`--danger-ink`	#6b1b2c	`#f6b8cf`	Red text
		`--danger-line`	#e46a9b	`#a8241f`	Red border
		`--info-soft`	#a9d6ea	`#1f4a6a`	Blue wash
		`--info-ink`	#1f4a6a	`#a9d6ea`	Blue text
		`--info-line`	#3f8fb0	`#3f8fb0`	Blue border
		`--purple`	#4e3a8a	`#a894dc`	Purple: Book, magic
		`--purple-soft`	#c3b2ea	`#3e2f66`	Purple wash
		`--purple-ink`	#3e2f66	`#c3b2ea`	Purple text
		`--focus`	#25637d	`#a9d6ea`	Keyboard focus ring

Stitched Toy chrome

Light	Dark	Token	Light hex	Dark hex	Use
		`--edge-lt`	#74462a	`#9a6b3f`	Frame: lit upper-left edge
		`--edge-dk`	#2a1a10	`#74462a`	Frame: dark lower-right edge
		`--drop`	#c9a96e	`#291d4a`	Frame: 2 px stepped drop
		`--ol`	#2a1a10	`#2a1a10`	Umber outline (buttons, felt bars)
		`--thread`	#e8a317	`#f5cf6a`	Stitch thread on cards
		`--thread-bright`	#f5cf6a	`#f5cf6a`	Hem stitch on felt
		`--felt`	#3d2a1c	`#3d2a1c`	Umber felt: header, bottom bar, footer
		`--felt-2`	#4a2c1a	`#4a2c1a`	Felt patches (header pills)
		`--felt-ink`	#f2e4c0	`#f2e4c0`	Text on felt
		`--felt-muted`	#c9a96e	`#c9a96e`	Quiet text on felt
		`--on-fill`	#f2e4c0	`#f2e4c0`	Text on solid fills

Chips and rarity

Light	Dark	Token	Light hex	Dark hex	Use
		`--chip-energy-bg`	#fff3c8	`#6a4510`	Energy chip
		`--chip-gold-bg`	#f5cf6a	`#6a4510`	Petals chip
		`--chip-lvl-bg`	#c8d09f	`#1f3f3a`	Level chip
		`--r-c`	#5b6470	`#5b6470`	Rarity: common
		`--r-u`	#2f5a2a	`#2f5a2a`	Rarity: uncommon
		`--r-r`	#25637d	`#25637d`	Rarity: rare
		`--r-l`	#7a4a1f	`#7a4a1f`	Rarity: legendary

Game art

Light	Dark	Token	Light hex	Dark hex	Use
		`--plot`	#6b4a2b	`#6b4a2b`	Garden soil
		`--sel`	#e8b923	`#e8b923`	Selected tile
		`--map-sea`	#a9d3e3	`#1e3a4e`	Region map sea
		`--map-road`	#6e4f2e	`#3f2b17`	Region map road

Area hues

Each part of the game tints its pages through `--area*` tokens set by `body[data-area]`.

Area	Light hue	Light wash	Dark hue	Dark wash
garden	#2f5a2a	#c8d09f	#7ab55c	#1f3f3a
world	#3f8fb0	#a9d6ea	#3f8fb0	#1f4a6a
kitchen	#e0501f	#ffc27a	#f08a24	#6b1b2c
book	#7a5cc0	#c3b2ea	#a894dc	#3e2f66
market	#b07a10	#fff3c8	#e8a317	#6a4510
community	#e46a9b	#f6b8cf	#e46a9b	#6b1b2c

Type, space and breakpoints

Type

System font stack, no web fonts: `system-ui, -apple-system, "Segoe UI", Roboto, Helvetica, Arial, sans-serif`. Body is 16 px on phones and 15 px from 721 px up.

Style	Size	Weight	Sample
h1	24 px (26 from 721)	800	**Fernhollow Patch**
h2	19 px	800	**Section heading**
h3, `.subhead` 15	16 px	800	**Sub heading**
body	16 px (15 on desktop)	400	Short lines, plain words.
choice title / sub	15 / 12.5 px	600 / 400	Plant carrots Ready in 3 days
small, help, chip	13 px	400	One short muted line
tabs, small button	14 px	500	Garden · Beds
code	13 px	mono	`choice_row()`

Space, shape, touch

Token	Value	Use
`--tap`	44 px	Minimum touch target (buttons, inputs, summary rows)
`--radius` / `--sm` / `--pill`	0	Hard pixel corners everywhere (v2.0.32). Frames notch the corners instead.
`--px`	2 px	One art pixel: frame edges, stitches, drops
`--frame` / `--frame-lite`	2 px + 2 px drop	Box-shadow frame for panels / one-tone frame for rows inside them. No layout space: leave ~4 px room
`--hem`, `--st-h`, `--st-v`	4 on / 2 off	Stitch lines (hard-stop stripes): felt hems / edge stitch
`--gut`	16 px (24 from 721 px)	Side gutter of the page
`--hdr`, `--bar`	52 (60) / 56 px	Slim top bar / phone bottom bar
`--maxw`	1392 px	Widest content
`.box` padding	14 × 16 px	Card inner space, 16 px below
gaps	10, 12, 14 px	Choice list and tabs 10, tiles 12, stat cards 14 (room for the frames)
`--shadow`	2 2 0, no blur	Hard pixel drop. Popups use `--shadow-2` (outline + 6 px drop)

Pixel frames (v2.0.32)

- No `border-radius` and no blurred shadows. A panel is framed with `box-shadow: var(--frame)`: 2 px edges, lighter on the lit upper-left (`--edge-lt`), darker lower-right (`--edge-dk`), a 2 px stepped `--drop`, notched corners. Rows inside a panel use `--frame-lite`.
- Frames take no layout space: give framed things about 4 px of room inside scrolling boxes.
- Buttons are felt toys: variants only set `--b-bg --b-hi --b-lo --b-ink --b-hover`. Margins are vertical only, so full-width buttons never overflow.
- Stitches are hard-stop stripes, 4 px on / 2 px off: `--hem` on felt, `--st-h` / `--st-v` for the edge stitch on special cards (prize, event, landing hero). Ordinary boxes stay unstitched.
- Pixel images use `.pix` (`image-rendering: pixelated`) at whole-number scales only. UI tiles live in `assets/img/ui/` (made by `tests/gen_ui_chrome.lua`, written at 2x).

Breakpoints

- **Phone first**: design for about 412 px wide, dark. Base CSS is the phone.
- **721 px and up**: tablet and desktop. Gutter 24 px, header 60 px.

- **1200 px and up**: groups in the menu stay open (`.navfixed`), two-column tiles. Content never exceeds 1392 px.
- No sideways scroll at any width. Wide things wrap, fold or scroll *inside* their own box.

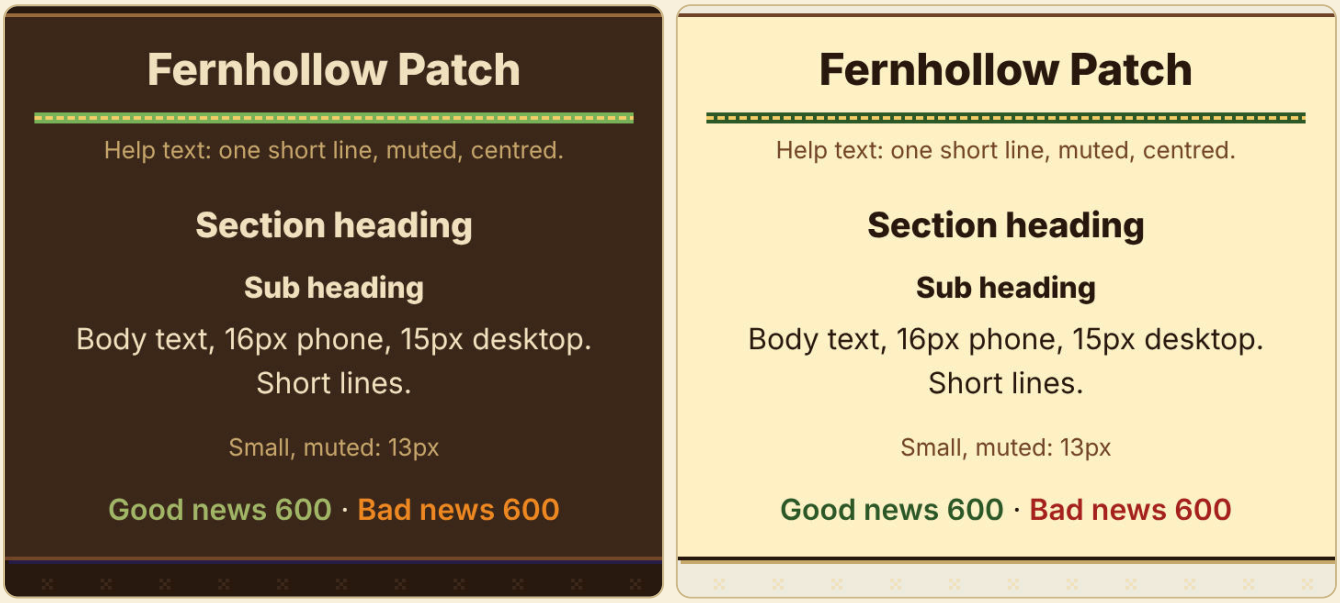
Components

Real markup and the real `style.css`, captured at build time. Each pair: dark (default) on the left, light on the right. Since v2.0.32 they show the Stitched Toy frames.

Type

Dark

Light

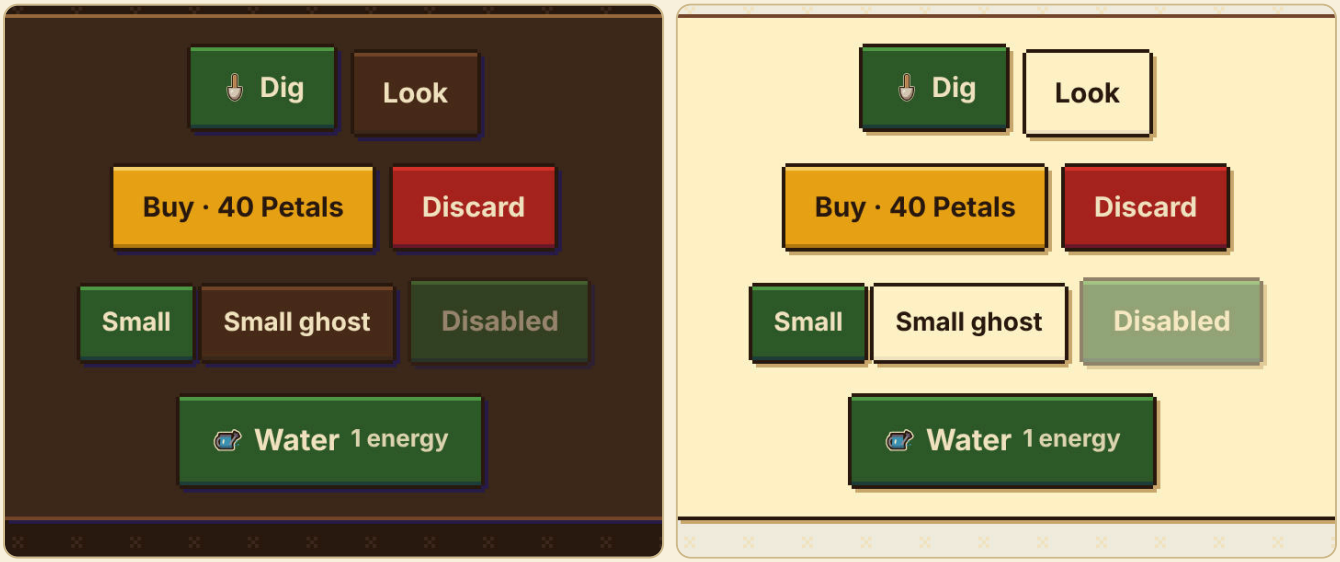


`.box` felt patch (`--frame`), centred: h1 24 (26 desktop) on an area ribbon with a gold seam, h2 19, h3 16, all 800; body 16, small 13, `.help` 13 muted.

Buttons

Dark

Light



`.btn` felt toy: lit top, shaded bottom, umber outline, presses 2 px. Solid moss = main verb. `.ghost` for the rest. `.gold` buys, `.red` destroys. `.small` 36 px, `.act` 48 px. Tap target 44 px.

Tabs

Dark



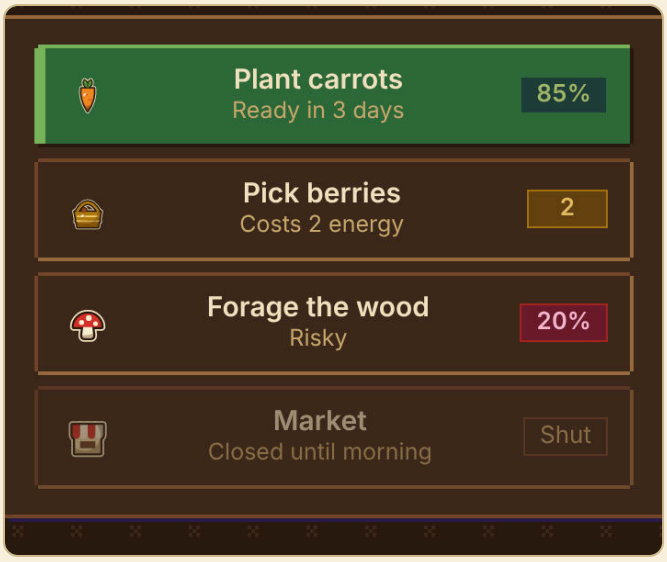
Light



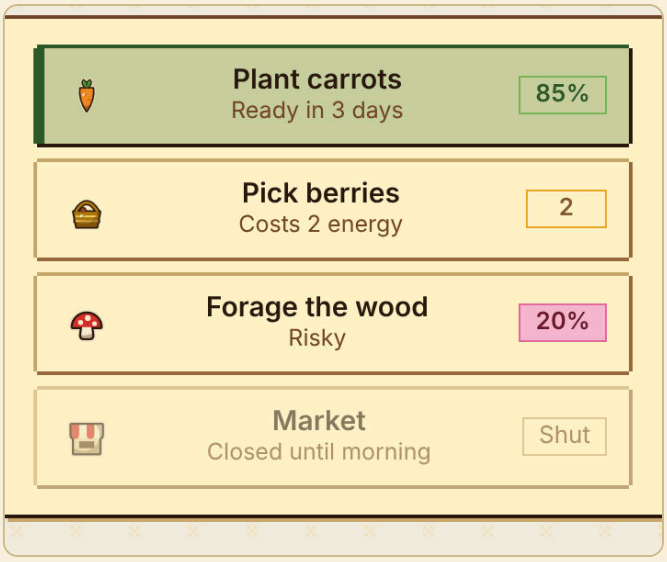
`nav.tabs` felt tags (on = solid area colour, outlined) and `.chip` status chips (`chip-in` , `chip-off` , `chip-pest` , `chip-rain`).

Choice

Dark



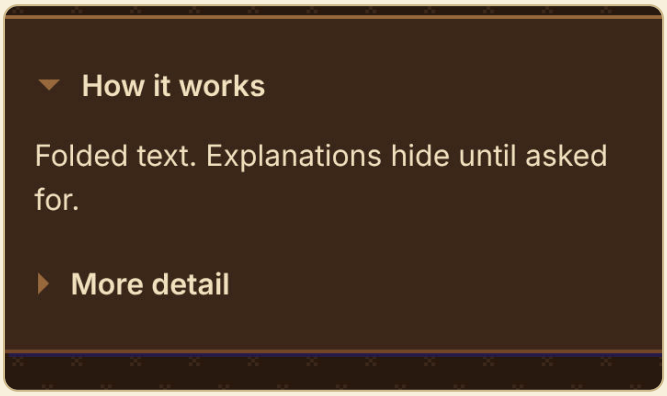
Light



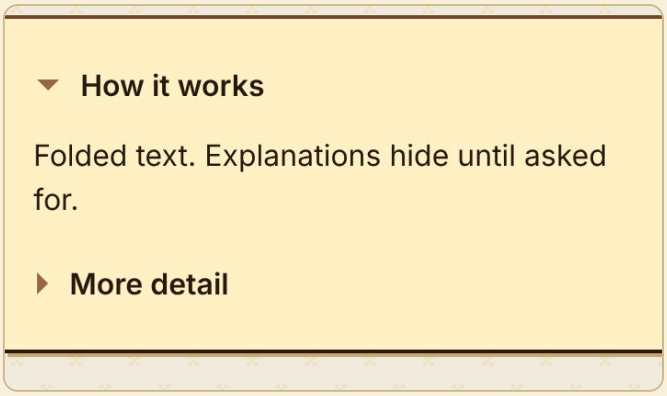
`choice_row()` rows: icon, title, small sub-line, badge (`good` , `cost` , `poor` , `free`). `primary` / `is-selected` = moss edge and inner band.

Fold

Dark



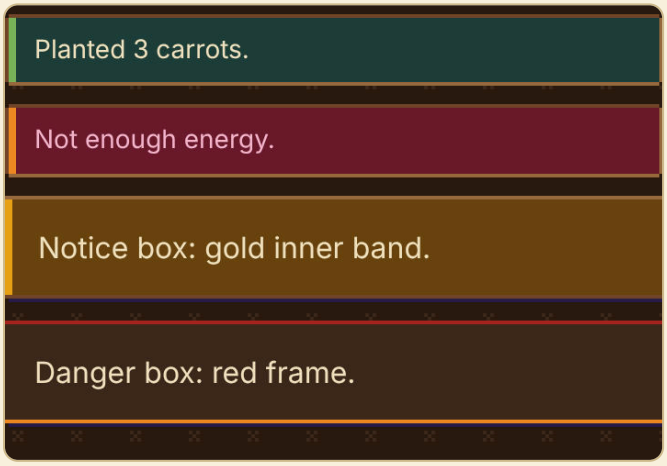
Light



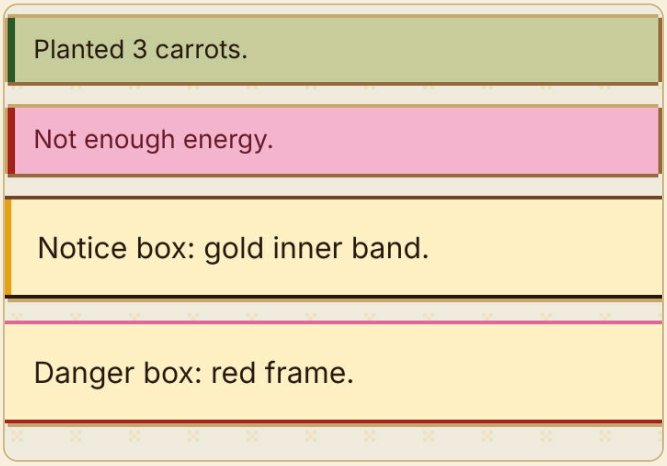
`details.fold` : closed by default. Explanations live here.

Flash

Dark



Light



`.flash` messages (ok, `err`) and `.box.notice` / `.box.danger`.

Stitch

Dark



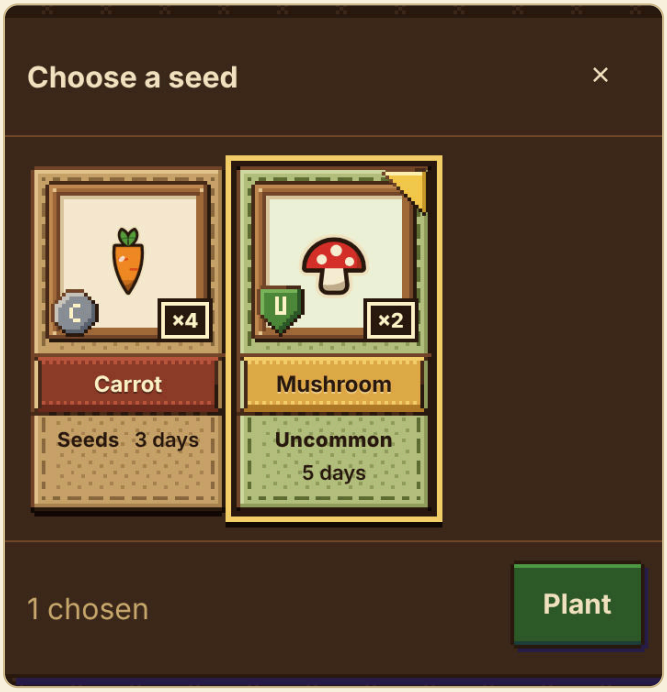
Light



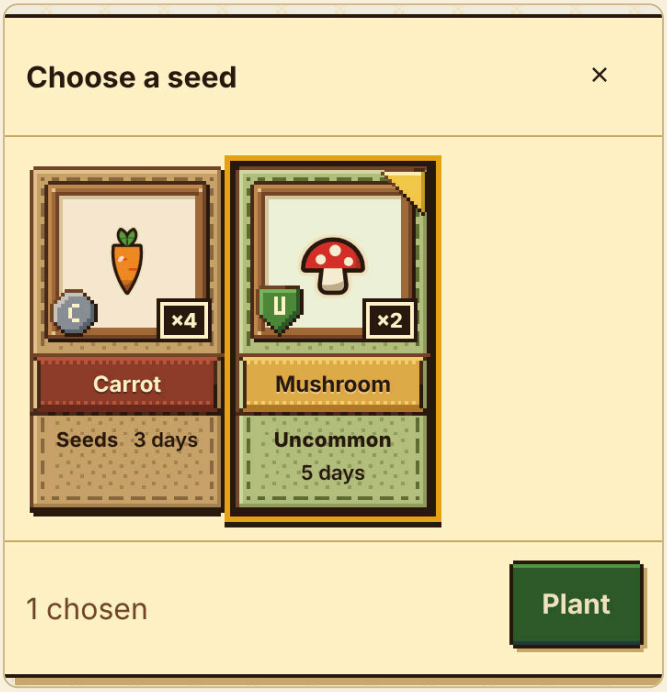
Special cards only: `.prizebox` / `.eventbox` get the gold edge stitch (`--st-h`, `--st-v`); `hr` is a stitched seam with a button knot.

Picker

Dark



Light



`dialog.picker` > `.pickbox` : head, a grid of item squares (`.itm` from `item_square()` , picked = `aria-pressed`), foot with one solid button.

Form

Dark



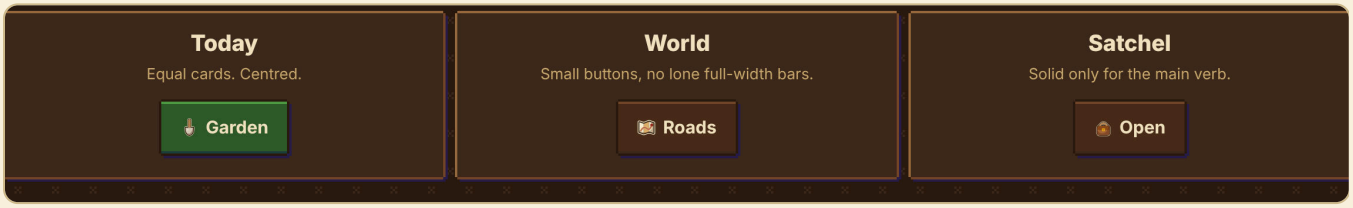
Light



`.form` labels above inputs, 44 px fields, 16 px text so phones do not zoom.

Desktop, 1200 px

Dark



Light



Equal cards, centred, small buttons. Shown scaled down; same rules from 721 px up.

Joshua's UI rules

These come straight from Joshua. They are checked on every screenshot before a release.

The rules

1. **Centre things**: text, headings, cards, buttons, result lines. Check every screenshot for it.
2. **No sideways scrolling.** Ever.
3. **Phone first.** About 412 px, dark theme. Desktop is the same design with more room.
4. **Fold explanations.** Rules, hints and long text hide inside `details.fold`.
5. **Few words.** Short labels, short lines. If it needs a paragraph, fold it.
6. **Solid buttons only for the main verb.** Everything else is `ghost`.
7. **No lone full-width button** inside a wide desktop card.
8. **Equal tile grids** for action buttons. Rows of choices use `choice_row()`.
9. **Many-option filters** are a `<select>` in a GET form (`data-autosubmit`), not a wall of chips.
10. Keep the top bar slim.

Build rules

- CSP is `style-src 'self'`: **no inline styles**. All CSS in `assets/style.css` (new work in the marked block at the end).
- Popups are `<dialog>` with `gvOpenDlg` / `gvCloseDlg`. Confirmations use `data-confirm` or `gvAsk`. Never `alert()` or `confirm()`. Never lock body scroll.
- Tap targets are at least 44 px (`--tap`). Small buttons are 36 px with a bigger hit area.
- In-place page loads need the page in `AJAX_PAGES`.
- Art falls back in order: real art, flower art, placeholder (`assets/img/ph/`), emoji.

Do and don't

Do

- ✓ Centre text, headings, cards, buttons
- ✓ Fold explanations
- ✓ Few words, short lines
- ✓ One solid button per card
- ✓ Tokens for every colour
- ✓ Dark first, then check light
- ✓ Equal tiles for actions
- ✓ `<dialog>` for popups

Don't

- ✗ Sideways scroll
- ✗ Inline styles (CSP blocks them)
- ✗ Hard-coded colours
- ✗ `@media (prefers-color-scheme)` for dark
- ✗ `alert()` / `confirm()`
- ✗ Lone full-width button on desktop
- ✗ Long text walls on the page
- ✗ Emoji in new UI (use an icon)

Checklist and upkeep

Before you ship UI

- Look at it on a 390 px dark screenshot. Then light. Then 1200 px.
- Is everything centred? Any sideways scroll? Any wall of text that should be a fold?
- Is there more than one solid button in a card?
- Did new text use an emoji? Use an icon (or a plain word) instead. Player text goes through `eu()`.
- Run `php -l`, the suites in `claude/handoff.md`, then rebuild this guide.

Keep the guide current

- Icons changed? Run `python3 tests/build_guide.py`.
- New component class or token? Add a demo to `tests/guide_src/components.html` and a line in `text.py`.
- New rule from Joshua? Add it to section C4 and to the do / don't lists.
- Commit the new PDF with the release so Admin can serve it.

Aseprite: Stitched Toy pixel art

Stitched Toy is the Aseprite (pixel) version of Hearthlight. It draws every sticker, icon, bust, scene and map tile as **hard-pixel art** that looks like a hand-stitched felt toy with enamel gloss. The game loads the PNG first and falls back to the SVG (`ph` , `ico` , `animals` , `landing` , `stops` , `gather`). Where Part D and Part A disagree, **Part D wins for pixel art**. Part A still rules palette families, names and the 64 grid.

> Since v2.0.34 stickers, icons and busts have **no cream halo**. They get a 1 px black border instead (see “Outline and border”). The Part A pages describe the SVG files, which still ship as the fallback and keep their halo.

The signature

- **Hard pixels only**: no anti-aliasing, no blends, no gradients, no dither. Every pixel is a solid palette colour or fully transparent.
- **Hue-shifted ramps**: shadows go warmer and deeper, lights go yellow-orange. Never just darken.
- **Stitched seam**: dashed thread gold `#f5cf6a` , 2 px on and 1 px off, wherever there is a real join (back seam, lid, hem).
- **Edge stitch**: a 1 px dashed gold ring just inside the body edge, about every third diagonal pixel.
- **Button**: a 4 × 4 cut-corner gold disc with a darker lower right and two thread holes, at the head of a seam. It must still read at 32 px.
- **Enamel gloss**: a short comma on the lit upper left (`#ff9158` , dot `#ffc27a`). Spots get a 1 px glint so they look glazed.
- **Eyes (locked, never redesign)**: 5 × 5 cream `#f2e4c0` disc with cut corners, 3 × 3 umber `#2a1a10` pupil, 1 px cream glint on the pupil’s upper-left pixel. Set wide on the head. No blush patches.
- **Shape language**: chibi, big head (about 5/6 of the body width), 2 px bent limbs in lavender `#54447f` with a lit top `#7b66b4` , 4 × 4 club tips on antennae, diamond or bean markings.

Canvas sizes

Kind	Canvas	Notes
Sticker, icon (`ico-*`)	64 × 64	At least 3 px clear before the border. Icons use chunky shapes, parts 3 px or wider
Bust (`npc-*`)	80 × 100	May run off the bottom edge (no border there). Eyes scale down only if the head is under 20 px wide
Map stops, small art	32 to 48	Scale pixel numbers by k = shorter side / 64, rounded. Outline stays 2 px, stitches and glints 1 px
Scenes, banners, landing	about 100 px short side, written at 2x	Do NOT scale the outline up. Nearest-neighbour 2x keeps line weights in proportion
Trek map tiles	42 × 42 (sheets 84 x 126)	No border, see Trek map tiles

Outline and border

- **No cream halo** (changed 2026-10-22). Everything outside the art is transparent.
- **Last step, 1 px black border**: every transparent pixel **4-adjacent** (plus-shaped, no diagonals) to a non-transparent pixel becomes opaque `#000000`. It goes round the outside and round interior holes. Enclosed cream pockets under 8 px (eyes, glints) are kept. Applies to **stickers, icons and busts**. **Scenes, banners, landing art and map tiles get no border.**
- **Umber outline under the border**: 2 px grown from the silhouette, plus 1 extra px on the lower right (where `s = (x - cx) + (y - cy)` is above 12). Never pure black.
- **Selective by light**: `s < -18` is `#74462a`, `s < -2` is `#4a2c1a`, everything else `#2a1a10`. A pale subject (dust, hair, paper, bone) gets an inside outline of `#4a2c1a` on all sides.
- **Inner contour**: 1 px `#2a1a10` where one big shape overlaps another.
- **Shading**: one crescent at the lower right, a 1 px bounce line inside its edge, a 1 px rim on the upper left, a 1 px contact shadow under an overlapping part. One light direction everywhere.
- Scenes and map objects keep the three-tone outline on raised objects, one crescent per object, and stepped bands (about 5) instead of gradients.

Palette

Use	Colours
Border (outside the art only) the old cream halo #f2e4c0 is retired	#000000
Outline, light to dark	#74462a #4a2c1a #2a1a10
Red ramp gloss, gloss dot, rim, base, bounce, shade, deep	#ff9158 #ffc27a #f2793a #dd3a2a #b8452f #ae2a2a #6b1b2c
Dusk (head) rim, base, shade, gloss	#7b66b4 #3e2f66 #291d4a #a894dc
Legs (base, lit top)	#54447f #7b66b4
Spots (base, glint)	#35161f #5e2a3a
Thread gold (stitch, shade, button holes)	#f5cf6a #e8a317 #8a4a1c
Eyes (disc, pupil)	#f2e4c0 #2a1a10

For other colours use the **ramp recipe**, starting from the Part A family colour as base: gloss = toward yellow-orange and about 35% lighter; shade = toward red-violet and about 22% darker; deep = further toward violet and about 45% darker; bounce = shade warmed and 8% lighter; rim = gloss 10% darker. Example: moss `#4f9a45` gives shade `#2f6a3a`, deep `#1f3f3a`. Pale subjects use the cream-safe set: gloss `#fff3c8`, dot `#ffffff`, rim `#f5b860`. Never use `#000000` or `#ffffff` except the border and one white dot on a pale subject.

Pixel samples and trek map tiles

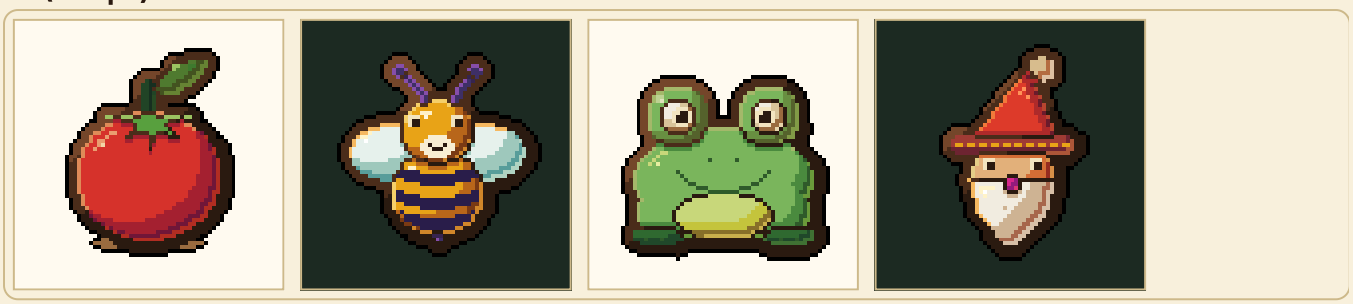
Real files from `game/assets/img`, drawn at whole-number scales with nearest-neighbour. Each strip is shown on a light and a dark chip so the black border is checked on both page themes.

Stickers, icons and animals

1x (64 px) on a light chip, then a dark chip



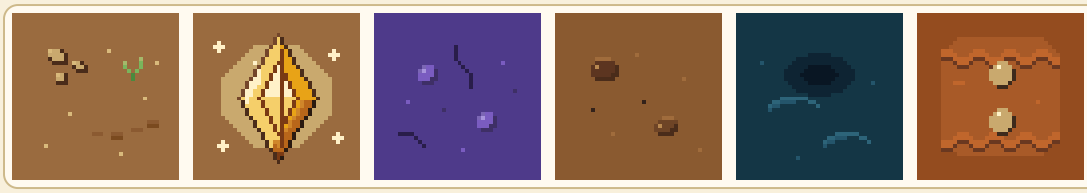
2x (128 px)



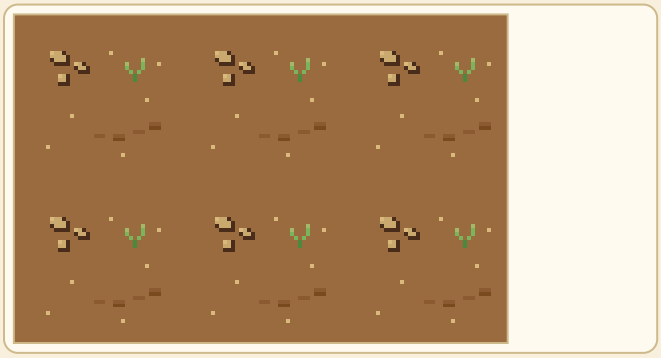
Tomato, bee, frog and gnome hat. The 1 px black border is the last pixel ring; the umber outline sits under it. Files: `ph/`, `ico/`, `animals/` PNGs.

Trek map tiles

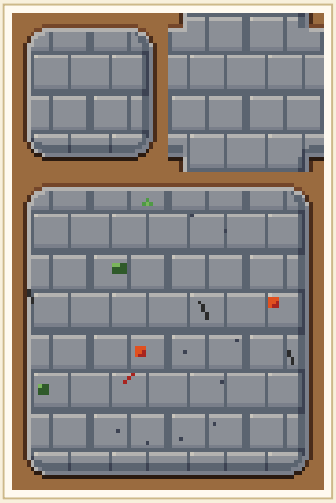
The trek map is drawn from **46 hand-made tiles** in `assets/img/trek/`. Set `TK_PIXEL_TILES` false (before `includes/trek_art.php` loads) for the old code-drawn map. `TK_PIXEL_ME` (default false) swaps the player's own gnome for the pixel `trek-me` marker. Tiles have **no halo and no border**; one shared palette serves open and underground.



42 × 42 tiles at 2x: floor-o-1, crystal-o, blight-o-1, floor-u-1, chasm-1, clay-wet.



floor-o-1 repeated 3 × 2 at 2x: no seam.



wall-o-auto, 84 × 126 at 2x: 4 × 6 quarters of 21 px.

Group	Tiles
Open ground (10)	`floor-o-1..3`, `mud-o-1..2`, `roots-o-1..2`, `blight-o-1..2`, `crystal-o`
Underground (7)	`floor-u-1..3`, `mud-u`, `roots-u`, `blight-u`, `crystal-u`
Water and clay (7)	`chasm-1..2`, `drained`, `clay-wet`, `clay-loose`, overlays `ripple-pool`, `ripple-high`
Auto-join and walls (9)	sheets `wall-o-auto`, `wall-u-auto`, `water-auto` (84 × 126); `wall-o-deep`, `wall-u-deep` (42 × 42); caps `wallcap-n/s` (42 × 5), `wallcap-e/w` (5 × 42)
Bridge and rails (6)	`bridge-h`, `bridge-v`; rails `rail-n/s` (42 × 4), `rail-e/w` (4 × 42)
Markers (7)	`token-gold`, `token-ember` (34), `me` (38 × 40), `foot-h`, `foot-v` (12), `bang` (24), `flag` (20)

All files are `trek-<name>.png`. Ground tiles are 42 × 42, opaque and full-bleed.

Seamless rules

- The outer **3 px ring** of a ground tile is exactly the terrain's ring colour (table below). Variants of one terrain (`floor-o-1..3`, `mud-o-1..2` ...) swap with no seam.

- Details stay at least 3 px from the edges unless designed to cross. A ground tile has **no crescent**: its texture is lit or flat with 1 px accents.
- Umber outline only on **raised objects** (crystal, planks, rails, stones, wall masses, markers). Ground, water and clay have none.
- Every asset and variant is drawn separately: no flip, rotate or recolour reuse. Alpha is 0 or 255 only. Must read at 42, 35 and 21 px.
- Map-key meaning is kept: ripples = high water, two dots = wet clay, cracks = loose clay, diamond = crystal, tuft = roots, dots = mud or blight.

Tile family	Ring colour (outer 3 px)
`trek-floor-o-*, crystal-o`	`#9a6b3f` so_base
`trek-mud-o-*, mud-u, drained, clay-loose`	`#7a4a1f` mud_base
`trek-roots-o-*`	`#7d8a62` or_base
`trek-blight-o-*`	`#4e3a8a` du_base
`trek-floor-u-*, crystal-u`	`#8a5a30` su_base
`trek-roots-u`	`#4a4632` ur_base
`trek-blight-u`	`#3e2f66` du_shade
`trek-chasm-*`	`#143645` ch_base
`trek-clay-wet`	`#944c1f` cl_base
`wall-o-auto empty side apron`	`#9a6b3f` floor-o ring
`wall-u-auto empty side apron`	`#8a5a30` floor-u ring
`water-auto empty side apron`	`#7a4a1f` mud ring

Auto-join sheets

`wall-o-auto`, `wall-u-auto` and `water-auto` are **84 × 126**: 4 columns x 6 rows of 21 × 21 quarters (RPG Maker A2 style). The game composes the 47 neighbour combinations from four quarters, one per quadrant, chosen by the 8 neighbours:

- both side neighbours are mass: the interior quarter, or the inner-corner quarter if the diagonal is empty;
- one side is mass: the edge quarter on the empty side;
- neither: the outer corner quarter.

Every quarter shares the same skeleton on its four border lines, so any two can sit side by side. On an empty side the first 3 px are the **apron** (the ground ring colour), then a 1 px outline (N `#74462a`, W and E `#4a2c1a`, S `#2a1a10`), a lit rim on N and W or a bounce line on S and E, then the base. The four iso quarters are pixel copies of the sheet's own corners.

Trek palette

Six steps per ramp, but one asset normally uses 8 to 14 colours. Gold thread (2 on, 1 off) only at real joins.

Ramp	gloss, rim, base, bounce, shade, deep
so (open soil)	#d9ba7c #c9a96e #9a6b3f #8b5a31 #7a4a1f #3d2a1c
su (underground soil)	#b88850 #a06c3c #8a5a30 #6e4426 #5c3520 #33221a
mud	#9a6b3f #8b5a31 #7a4a1f #6e4426 #5c3520 #3d2a1c
or (open roots)	#b9c58a #a2b076 #7d8a62 #6f7b56 #5f6c4a #3a4530
ur (underground roots)	#8a8a5c #6e6e48 #4a4632 #403e2c #38352a #272219
mo (moss)	#c8d77a #a4b86a #7ab55c #6a9a3a #4a8a3e #2f5a2a
sn (stone)	#c9c5bc #b3b2b0 #8b8f96 #7a7f8a #5b6470 #3c4252
wl (clay rock)	#e6cf98 #d9bd84 #c9a96e #b08a55 #9a6b3f #7a4a1f
du (blight)	#a894dc #7b66b4 #4e3a8a #463780 #3e2f66 #291d4a
wa (water)	#c8ecdc #a9d6ea #3f8fb0 #2f7fa0 #25637d #1c3a5e
ch (chasm)	#2d6479 #24556b #143645 #112f3d #0e2433 #091623
cl (wet clay)	#c0662c #a85a28 #944c1f #84451f #6a3820 #3d2a1c
wd (wood)	#ecc58a #d9a660 #b07a45 #a8582f #8a4f2a #5e2f1e
gd (gold, crystal)	#fff1c8 #f5cf6a #e8a317 #d0862a #b8661a #7a3a14
rd (red, ember)	#ff9158 #f2793a #dd3a2a #b8452f #ae2a2a #6b1b2c
skin	#ffe2a8 #f9d49c #e9b98a #d9885a #c47a52 #8a4234
cr (cream)	#fff3c8 #f5b860 #f4f0e0 #e2d6b8 #cfc3a2 #a89a78

Outline on raised objects: `#74462a` lit side, `#4a2c1a` middle, `#2a1a10` shaded side. Thread `#f5cf6a` / `#e8a317`, button holes `#8a4a1c`. Game colours kept as is: soil `#3d2a1c` `#7a4a1f` `#9a6b3f` `#c9a96e`, moss `#4f9a45` `#2f5a2a`, ink `#2b2b2b`. Never `#000000` or `#ffffff` in tiles.

Aseprite workflow and checklist

Where the source lives

`aseprite-source/` in the repo holds the Lua scripts and `.aseprite` files behind every PNG, the `gvlib` helper library with its brief, and the rule files. Edit a script and rerun it; never retouch a PNG by hand. Keep the `.lua` with the art.

Build one asset

1. Write a Lua script. Run it in batch mode: `"C:\Program Files\Aseprite\Aseprite.exe" -b --script my.lua` (Aseprite 1.3.18).
2. `local GV = dofile(".../gvlib/gvlib.lua")` then `local c = GV.canvas{ w = 64, h = 64, mode = "sticker" }`.
3. Describe the shapes and colours; `c:build{...}` paints back to front. Parts get the base, crescent, bounce, rim, gloss and contours for free.
4. Add details after the parts: `c:seam`, `c:button`, `c:eye`, `c:spot`, `c:paint`. Draw a part only once.
5. `c:save("x.aseprite", "x.png")` builds the outline, saves both files and returns the check numbers.

gvlib mode	Canvas	Border	Use
`sticker`	64 × 64	yes	Default: pictures, creatures, items
`icon`	64 × 64	yes	`ico-*`: chunky parts, longer gloss
`bust`	80 × 100	yes (not at the bottom edge)	`npc-*` folk portraits
`scene`	any, `out=2` for 2x	none	Banners, landing art, backgrounds, stops

> The shipped art got its black border from a post-pass (`batch.py`). Check whether your copy of `gvlib.lua` still draws the old halo pens before drawing new stickers; if so, add the border pass.

Check scripts

- `GV.check` (run by `c:save`): size, margin, `semi` (must be 0), colour count, `bw` (pure black or white pixels: 0 unless the border or a pale dot).
- `trek-pixel/check.py`: reads `palette.md` and warns on any opaque pixel not in the palette, and above 16 colours.
- View the PNG at 64, 32 and 16 px (tiles at 42, 35 and 21).

Checklist for every asset

- Every pixel solid (alpha 0 or 255), no blends, dither or gradients?
- Stickers, icons, busts: 1 px `#000000` border on every outside edge (4-adjacent), no cream halo left? Scenes and tiles: no border?
- 3 px of room inside the frame (64 px canvas) before the border?
- One crescent, the bounce line and the 1 px contact shadow, and no other shade shapes?
- Outline selective by light, never pure black under the border?
- Eyes exactly as locked? Seam, edge stitch and button where the object has a join?
- Only palette colours (Part A, the Part D table, or the ramp recipe)?
- Limbs and shafts at least 2 px; stitches and glints 1 px?
- Readable at 64, 32 and 16 px?
- Name matches the original file name exactly (`<folder>/<name>.png`)? `broken-*` is the intact drawing plus damage; `stage-*` grows one feature per step.
- Avoid: dither bands, blush or cheek patches, zipper or double-wide stitching, blobby markings, shading under 2 px limbs.

Scene art addendum: Aseprite and black border rules

Rules for the **scene overhaul** art (stage backdrops, banners, maps, interiors, fight effects, battle sprites). It adds to Part D and never replaces it. Where this page and Part D disagree, this page wins for the files named here. The file list, names and counts are in `claude/art-spec-overhaul.md`.

Which pieces get the black border

The 1 px `#000000` border is for things that **stand on a scene**. A scene itself has none. The page background and the stage are never bordered.

Piece	Border	Mode (gvlib)	Notes
Stage backdrops, banners, maps, interiors	**none**	`scene`, `out = 2`	Three-tone umber outline on raised objects only
Stage miniatures (party, crew, enemies, bosses): see Toy miniatures	**yes**, all sides	`sticker`	Transparent outside, border last. Face right. Sit on a round base
Villager busts (`npc-*`, `npc-unknown`)	**yes**, not on the bottom edge	`bust`	Stand behind a counter or half-wall so the cut bottom is hidden
Fight effects (`fx-*`)	**none**	`scene`	Short-lived, drawn over a scene. 1 px umber contour only where a shape has an edge
Region mask (`map-world-regions`)	**none**	raw	Flat colours, no outline, no stitches. Never resized

Black border rules (restated for this set)

- Last step only: every transparent pixel **4-adjacent** (plus-shaped) to an opaque pixel becomes `#000000`. Round the outside and round interior holes. No diagonals.
- It is added after the umber outline. The umber outline stays under it (2 px, +1 on the lower right). Never use black for the outline, shading or detail.
- Border is **1 px at any canvas size**. Do not scale it with k. A 96 × 96 boss still gets a 1 px border.
- No cream halo on anything in this set.
- Do not border a sprite twice. Border once, on the final cropped sprite, before saving the PNG.

- Check on a light and a dark chip. If the border vanishes on dark, the umber under it is too dark: lighten the lower-right outline tone one step.
- Pure black `#000000` is allowed **only** for this border. A black pixel anywhere else is a fail.

Aseprite rules for this set

- **Aseprite 1.3.18**, batch mode, driven by a Lua script that uses `gvlib`. Never retouch a PNG by hand. Keep the `.lua` and `.aseprite` with every PNG in `aseprite-source/final/aseprite/scenes/` (scenes) or the matching existing folder (sprites).
- **Colour mode: RGB**, 8 bits, with a **palette limit** per file (see the table). No indexed-colour tricks that change colours on export.
- **Hard pixels**: alpha is 0 or 255 only. No blur, smooth, outline-by-brush, gradient tool, dither pattern or layer opacity below 100 on the exported result. Layer blend mode: Normal only.
- **Layers** (top to bottom, names exactly): `border` (sprites only), `fx`, `detail`, `objects`, `ground`, `far`, `sky`. Merge nothing before the last step so the file stays editable. Hidden helper layers go in a group named `guide` and are never exported.
- **One frame** per file unless the spec says sheet. Sheets are **horizontal strips**, frames side by side with no gap, no padding, all the same size. Do not use animation tags for the engine; the game reads the strip by frame size.
- **Scenes are drawn small and exported at 2x**: backdrops on a 240 × 135 grid, banners 240 × 60, world map 320 × 180, region maps 240 × 135. Export with **Nearest-neighbour 200%**. This keeps line weights in proportion; do not draw the outline at the exported size.
- **Export**: `File > Export` equivalent in the script, `.png`, no trim, no scale beyond the 2x above, no metadata, 32-bit RGBA. Size must match the spec exactly or the file is rejected.
- **Naming**: lower case, hyphens, exactly the names in the spec. No spaces, no version suffix. Replace a file by overwriting; do not make `-v2`.
- **Transparency**: sprites and effects have a transparent background. Scenes are fully opaque (alpha 255 everywhere).

File type	Grid drawn	Exported	Max colours	Border
Stage backdrop	240 × 135	480 × 270	32	none
Banner	240 × 60	480 × 120	24	none
World map	320 × 180	640 × 360	40	none
Region map	240 × 135	480 × 270	32	none
Interior	240 × 135	480 × 270	32	none
Miniatures	see Toy miniatures	same	see that table	1 px
Bust	80 × 100	80 × 100	16	1 px, not bottom
Fight effect	per spec	per spec	8	none

Scene composition rules

- **Ground line**: backdrops put the walkable ground between y 190 and 215 (exported pixels). Characters stand with their feet on it. Keep the left and right thirds calm and the centre readable.
- **Depth by bands**: about 5 hard-edged colour bands for sky and haze, then far, mid and near layers. Each farther layer is lighter and less saturated. No gradients, no dither.
- **Raised objects** (trees, rocks, furniture, stalls) get the three-tone umber outline, one crescent, a bounce line and a rim. Ground, water, sky and haze get **no outline**.
- **Light**: always from the upper left. Windows and fires in interiors may add a warm rim, but the crescent stays lower right.
- **Stitched-toy signature in scenes**: a thread-gold hem stitch (2 on, 1 off) along the top and bottom edge of **banners** only. Seams and buttons only on real joins (a quilt, a sack, a banner cloth). Do not stitch terrain.
- **No text, no UI, no characters** in backdrops. Small background life (a bird, a lantern flame) is allowed and must be at least 4 px.
- **Cosy not scary**: even the lair, delve and Trek boss scenes stay warm in value. Danger is shown by shape and colour temperature, never gore or pure dark.
- **Palette**: only Part A families, the Part D table, or the ramp recipe. State which family you borrowed for a new landscape. Day lighting only (night is a code tint).
- **Maps**: draw regions as clearly different colour families so they read at 25 percent size. Leave a flat 24 × 24 spot for each pin (world) or 20 × 20 (region). Roads and mycelium paths 3 px wide minimum.

Checks before a scene file ships

- Exact size, 32-bit RGBA, alpha 255 everywhere (scenes) or 0/255 only (sprites)?
- Under the colour cap, every colour from the palette rules, `bw` check shows no black except the sprite border?
- Scenes: no border, no halo, no gradient, no dither, no text?
- Sprites: 1 px black border on every outside edge, none on the bottom edge of a bust, facing right, feet on the bottom row?
- Ground line in the 190 to 215 band, side thirds calm?
- Viewed at 100 percent and at 50 percent: the subject still reads?
- Source `.lua` and `.aseprite` saved next to the PNG with the same base name?

Toy miniatures (stage characters)

Fight and stage characters are drawn as **toy miniatures**: little figurines in a 3/4 view standing on a round base, as if set on a diorama table. This fits Stitched Toy (felt and enamel toys). It is Aseprite-only pixel art: hard pixels, palette ramps, 1 px black border.

How to build one

Aseprite only. No 3D software, no voxel tools, no renders, no AI image tools, no imported images. Every pixel is placed in Aseprite (Lua script with `gvlib`, or by hand in the editor). Fake the 3D by hand: draw the figure in 3/4 view as a small solid with a lit top, a lit left face and a shaded right face. One crescent still falls lower right. A miniature that looks traced from a render or a photo is rejected.

Camera and pose

- View: 3/4 from above, elevation about 30 degrees, figure turned so it **faces right and slightly toward the viewer**. Same camera for every miniature so they sit together on a stage. Code flips the sprite for the other side.
- Pose: standing idle, weight even, weapon or item held low. One neutral pose per file.
- Optional extra frames only as a horizontal strip: `-idle` (2 frames), `-hit` (1), `-ko` (1, lying or sitting on its base).
- Feet and base sit on the bottom rows. The base counts as part of the figure.

The base

- A round disc seen as an ellipse, 2/3 of the sprite width, 6 px thick (top face, side face).
- Party bases are felt moss `#4f9a45` ramp with a thread-gold edge stitch. Enemy bases are felt soil or stone ramp with no stitching (they are not ours). Bosses get a wider base with a seam and a button.
- The base replaces the ground-shadow ellipse. No separate shadow.
- Gold stitch rule still applies: thread gold only on joins and base edges.

Proportions and detail

- Chibi: head about 5/6 of the body width, squat body, **locked eyes** (Part D). On a miniature under 24 px tall the eyes scale to 3 × 3 disc, 2 × 2 pupil.
- Limbs at least 2 px. Accessories (hat, beard, pack, tool) are separate readable shapes.
- Silhouette must read at 32 px. Every enemy has one clear feature (horns, shell, wings) in the outline.
- Shading: 3 value planes (top lit, left mid, right shade) per big shape, plus the one crescent, bounce line and rim. No extra shade shapes, no dither.

Sizes and counts

Sprite	Canvas	Max colours	Border
Gnome, crew, villager (battle)	64 x 80	20	1 px, all sides (base is at the bottom, so the bottom gets it too)
Small enemy	56 x 64	16	1 px
Large enemy	80 x 80	20	1 px
Boss	112 x 112	24	1 px

Existing 64 × 64 creature stickers stay for lists, the Almanac and icons. On the stage the code uses the miniature if the file exists, else the sticker standing on a code-drawn base.